

Reviewer Pack — Minimal Order of Twisted Derivative Modules and Circuit Mono...

Entry dc89358e260a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Twisted Derivative Modules and Circuit Monotone Width Inequality

Entry ID: dc89358e260a **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 14:03:32 UTC

1. Conjecture

Field A (mathematical branch): Algebra (Twisted Derivative Modules) **Field B** (complexity object): Boolean Circuit Complexity (Circuit Monotone Width)

Statement:

For every Boolean circuit C with n inputs and m outputs, the minimal order of a twisted derivative module $M(C)$ associated with C satisfies $|M(C)| = \Theta(m^{(2/3)n}(1/3))$.

Rationale (proposer's reasoning):

Twisted derivative modules provide a framework for studying algebraic structures in the context of computational complexity. The conjecture suggests that the order of such modules could serve as a lower bound on circuit monotone width, which is an important measure of circuit complexity. This bridge might expose new structural insights into both algebraic and complexity-theoretic properties.

Taxonomy category: TwistedDerivativeModule (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): a1750951cbd243c2

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, across at least 90% of the 30 seeds tested, the computed minimal order $|M(C)|$ for each circuit C with n inputs and m outputs lies within a factor of 2 from the expected value of $\Theta(m^{(2/3)n}(1/3))$, as measured by the aggregate mean.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "minimal order twisted derivative modules" AND "circuit monotone width inequality" - "Algebra" AND "Boolean circuit complexity" AND twisted derivative modules - "Twisted Derivative Modules" related to "Circuit Monotone Width"

Top relevant hits considered: - [http://arxiv.org/abs/cs/0608100v1] Similarity of Semantic Relations - [http://arxiv.org/abs/0810.1207v1] A Layered Grammar Model: Using Tree-Adjoining Grammars to Build a Common Syntactic Kernel for Related Dialects - [http://arxiv.org/abs/1310.8154v3] Characteristic cohomology of the infinitesimal period relation

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def generate_random_circuit(n, m):
    if n <= 0 or m <= 0:
        return None
    circuit = []
    for _ in range(m):
        gate_type = random.randint(1, 3) # 1: AND, 2: OR, 3: NOT
        if gate_type == 3:
            inputs = [random.randint(0, 1)]
        else:
            inputs = [random.randint(0, 1) for _ in range(gate_type - 1)]
        circuit.append((gate_type, inputs))
    return circuit

def evaluate_circuit(circuit):
    if not circuit:
        return None
    stack = []
    for gate_type, inputs in reversed(circuit):
        if gate_type == 3: # NOT
            result = 1 - inputs[0]
        else:
            if len(inputs) != gate_type - 1:
                return None
            if gate_type == 1: # AND
                result = all(inputs)
```

```

        elif gate_type == 2: # OR
            result = any(inputs)
        else:
            return None
    stack.append(result)
return stack.pop()

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    m_values = [n // 2 for n in n_values]

    results = []
    for n, m in zip(n_values, m_values):
        if n * m == 0:
            continue
        circuit = generate_random_circuit(n, m)
        if not circuit:
            continue
        result = evaluate_circuit(circuit)
        if result is None:
            continue
        results.append(result)

    if not results:
        return {
            "metric_name": "min_order",
            "metric_value": 0,
            "instances_tested": 0,
            "n_max": 0,
            "conjecture_holds": False,
            "counterexample": "empty_circuit"
        }

    min_order = len(results)
    expected_bound = Fraction(m ** (2/3) * n ** (1/3)).limit_denominator()
    support_fraction = sum(1 for result in results if abs(result - expected_bound) <= 2 * expected_bound)

    return {
        "metric_name": "min_order",
        "metric_value": min_order,
        "instances_tested": len(results),
        "n_max": max(n_values),
        "conjecture_holds": support_fraction >= Fraction(9, 10),
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

```


Safety rail: critic_challenge_falsified | original: The test results indicate that the conjecture does not hold for at least one seed, as evidenced by the counterexample provided. | next: Further investigation is needed to determine the correct definition of 'min_order' and to develop a more comprehensive testing strategy that covers a wider range of circuit sizes and configurations.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14169
2	preregistration	ollama_remote	glm4:latest	0	0	9992
3	novelty	ollama_remote	glm4:latest	0	0	8621
4	novelty	ollama_remote	glm4:latest	0	0	15249
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15393
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	42071
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13275
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12637
9	critic	ollama_remote	glm4:latest	0	0	54356
10	judge	ollama_remote	glm4:latest	0	0	9472

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 195236 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/dc89358e260a.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/dc89358e260a.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/dc89358e260a.tar.gz* (if generated)